

WASP-50b

R-1453

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_211022 · created 2026-08-01T09:45:20+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459509.856 +/- 0.058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.05
Transit depth (Rp/Rs)^2	2.77 +/- 1.68 [%]
Orbital Inclination (inc)	88.87 +/- 2.87
Airmass coefficient 1 (a1)	10.72 +/- 3.73
Airmass coefficient 2 (a2)	-1.72 +/- 1.36
Scatter in the residuals of the lightcurve fit is	33.85 %
Best Comparison Star	#3 - [27, 139]
Optimal Aperture	5.42
Optimal Annulus	13.58
Transit Duration (day)	0.04 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.05 vs 0.1404 (deviation 0.51σ)
Duration fit vs published	0.04 d vs 0.0754 d (deviation 1.22σ)
Mid-time O–C	0.8 min
Depth SNR	3.3
Residual scatter	33.85 %

Light curve

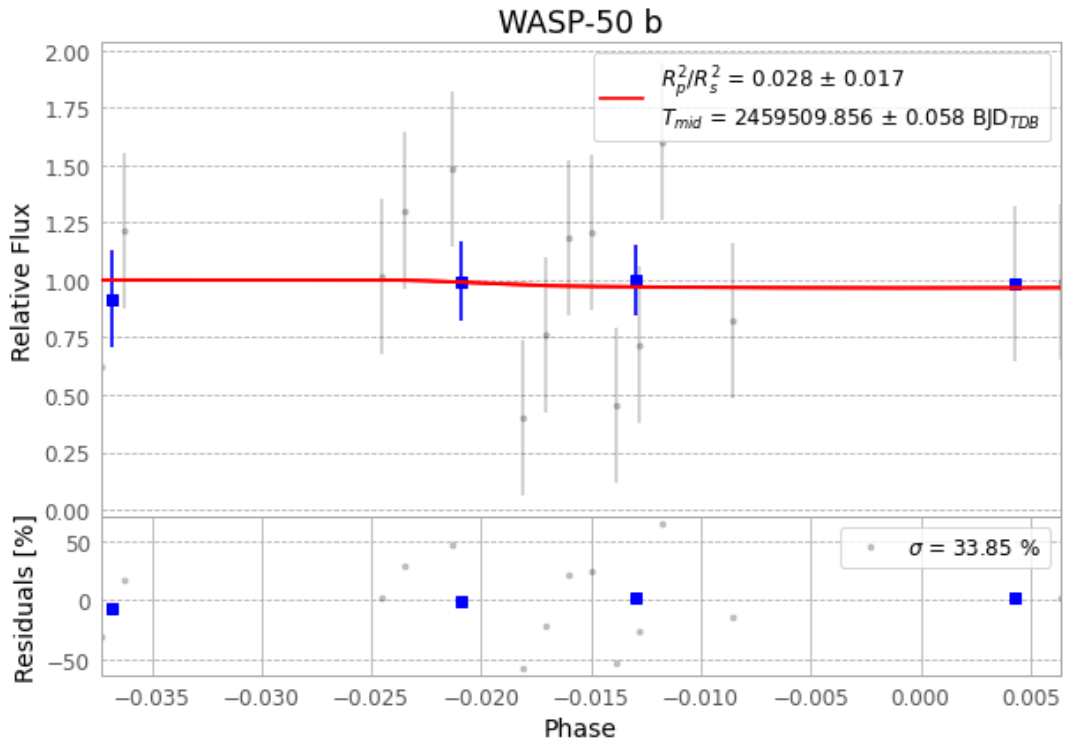


Figure 1 — FinalLightCurve_WASP-50 b_2021-10-21.png (SHA-256 75c5731dfdc5b51b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [465, 406], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 23ad03af196c564b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-50211022063612.FITS → WASP-50211022102112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-211022.log (891575621fe7...), FinalLightCurve_WASP-50 b_2021-10-21.png (75c5731dfdc5...), FinalLightCurve_WASP-50 b_2021-10-21.csv (67c368123f08...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 09:45Z.